

SUMMARY

AI Engineer specializing in production **LLM applications**, **RAG workflows** and **agentic AI systems**. Strong expertise in AI orchestration, retrieval systems, vector databases, API integration, and deploying scalable enterprise AI solutions using FastAPI, LangChain, LangGraph, OpenAI

WORK EXPERIENCE

KareXpert

Sep 2025 – Present

AI Engineer

- Developed backend services and REST APIs using Python and FastAPI for KareXpert's AI-Powered Business Intelligence platform, enabling self-service dashboard creation and embedded analytics while replacing Microsoft Power BI and eliminating recurring licensing costs of approximately ₹70K–80K per month.
- Designed and implemented an AI-powered Chart Builder in BI Tool using the Gemini API, engineering prompt workflows, LLM response validation, and backend orchestration to generate interactive visualizations from natural-language prompts, reducing dashboard creation time by 75%.
- Built a Retrieval-Augmented Generation (RAG) chatbot using Python, FastAPI, Qdrant, and OpenAI APIs, implementing semantic retrieval, prompt engineering, and response orchestration to enable conversational access to project tasks, documentation, and workflow insights.

1Gen

Aug 2023 – Oct 2023

Data Science Intern

- Leveraged Python, SQL, and machine learning techniques (regression and clustering) to analyze 15,000+ electronic health records (EHRs), identifying patient and operational trends that informed personalized care initiatives, while automating EHR data processing workflows to improve operational efficiency by 30% and developing Tableau dashboards to monitor clinical and operational KPIs.

PROJECTS

AskQL – Multi-Tenant Text-to-SQL Analytics Platform

Jul 2026

FastAPI, React, OpenAI GPT-4o, PostgreSQL, Google Cloud Run, Google Cloud Storage

[Ask QL](#)

- Architected a **multi-tenant Text-to-SQL** platform that converts natural-language queries into validated SQL with real-time token streaming and auto-generated chart visualizations. Implemented **AI guardrails** using an AST-based SQL validator to prevent prompt injection and unsafe query execution, achieving **100% accuracy** across a 30-case evaluation suite, with a hybrid PostgreSQL + GCS-persisted SQLite storage layer deployed on Vercel and Google Cloud Run.

StartupAI – Multi-Agent Startup Blueprint Generator [Blog](#)

May 2026

FastAPI, React, Groq (Llama 3.3-70B), Supabase, Vercel, Render

[Startup AI](#)

- Developed an end-to-end **multi-agent AI platform** that automates startup ideation by orchestrating a **7-agent sequential workflow** for market research, competitor analysis, revenue modeling, MVP planning, business synthesis, investor pitch generation, and PPTX export. Integrated FastAPI, Groq Llama 3.3-70B, Google OAuth (Supabase), and cloud deployment on Vercel/Render, enabling users to generate investor-ready startup blueprints in a single run while onboarding **15 users** within weeks of launch.

ReflexRAG – Self-Reflective Clinical Research Assistant

Apr 2026

FastAPI, Next.js, OpenAI GPT-4o mini, FAISS, BM25, AWS EC2, Nginx

[Reflex Rag](#)

- Built a **self-reflective RAG** pipeline for clinical document QA featuring hybrid FAISS+BM25 retrieval with Reciprocal Rank Fusion, HyDE embeddings, cross-encoder reranking, query expansion and an iterative grading loop that detects unfaithful answers and retriggers retrieval with wider context. Reduced repeated query **latency by ~95%** (20s to under 100ms) via Redis semantic caching (cosine sim ≥ 0.92) across 200+ chunks per document, evaluated using a **15-question GPT-4o-mini judge** harness and deployed on AWS EC2 with Nginx, PostgreSQL RDS, S3 and Celery workers for async PDF ingestion.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL

GenAI & Agentic AI: RAG, agent frameworks (LangChain, LangGraph, CrewAI), Embeddings, LLM APIs (OpenAI GPT, Claude), Hugging Face Transformer, Prompt Engineering, MCP, Fine-Tuning (LoRA, QLoRA), LLM Evaluation

Frameworks/Libraries: FastAPI, PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas

Databases & Visualization: MySQL, MongoDB, Neo4j, Redis, Chroma, FAISS, Tableau, Power BI

DevOps & Cloud: Linux, Git, GitLab, Docker, Kubernetes, AWS

EDUCATION

Boston University

Sep 2023 – Jan 2025

Master of Science, Applied Data Analytics

Boston, MA

Course Work: Deep Learning, Data Structures & Algorithms, Machine Learning & AI, Database Management System

ACHIEVEMENTS

- Secured **3rd place** in the AI Hackathon organized by the Machine Learning Department at Boston University for building an AI-powered clinical decision support prototype.
- Selected as a **Graduate Research Assistant** at Boston University to develop a multi-agent Clinical Research Assistant using LangGraph, hybrid RAG, FastAPI, and Azure under faculty mentorship.